SYSTEMATIC SEARCH FOR SINGULARITIES

SYSTEMATIC SEARCH FOR SINGULARITIES ~~IN MODELS OF 3D EULER-VOIGT FLOWS~~

BY

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Lay Abstract

A lay abstract of not more 150 words must be included explaining the key goals and contributions of the thesis in lay terms that is accessible to the general public.

The Navier-Stokes equations model the behaviour of viscous incompressible fluids. The question of whether or not the Navier-Stokes equations, or the inviscid Euler equations, are a valid description of fluid mechanics remains unanswered. A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. If the equations form a singularity from smooth initial conditions, then they are not valid descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to not form singularities from smooth initial conditions. However, by decreasing the regularization parameter we approximate Euler flows. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify smooth initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

Abstract

Abstract here (no more than 300 words). An abstract of not more than 300 words must be included and indicates the major emphasis of the thesis, new discoveries, and its contribution to knowledge. The style of abstract varies from discipline to discipline but the student should follow an appropriate style. This page must be numbered 'iv'.

The Navier-Stokes equations are a system of partial differential equations used to model the behaviour of viscous incompressible fluids. The question of whether or not unique classical solutions of the Navier-Stokes equations, and inviscid Euler equations, exist for all time, given smooth initial conditions, remains unanswered for the three-dimensional problem.

A singularity is the occurrence of an infinite derivative of the velocity at a single point in the fluid flow. The formation of a singularity from smooth initial conditions would invalidate the equations as accurate descriptions of fluid mechanics. The Euler-Voigt equations are an inviscid regularization of the Euler equations which are known to be globally well-posed from smooth initial conditions. However, by decreasing the regularization parameter the solutions of the Euler-Voigt equations converge

to those of the Euler equations. We examine the Euler-Voigt equations with a PDE-constrained optimization problem to identify smooth initial conditions that produce 'extreme' behaviour in numerical simulations. ## We present numerical evidence that CONCLUSION ##

Your Dedication

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Acknowledgements

Acknowledgements go here. An expression of thanks for assistance given by the Supervisor of research and by others should be either set forth on a separate page or incorporated into the Preface (if there is one). These and all subsequent preliminary pages listed under (f), (g) and (h) must be numbered in lower case Roman numerals, i.e., ‘iv’, ‘v’, ‘vi’ etc.

Dr. Bartosz Protas, Dr. Xinyu Zhao, Dr. Jörn Davidsen, members of committee, Parents

I would like to thank Dr. Bartosz Protas for his support and guidance on my research project. I also want to thank Dr. Jörn Davidsen of the Complexity Science Group for supervising my undergraduate research and inspiring me to pursue computational physics.

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Notation, Definitions, and Abbreviations

Notation

$A \leq B$	A is less than or equal to B
x	Plain-text to denote scalar x
$\boldsymbol{x}$	Bold-text to denote vector x
$\boldsymbol{x}^\intercal$	Transpose of vector x
I	Identity Operator

Definitions

Challenge With respect to video games, a challenge is a set of goals presented to the player that they are tasks with completing; challenges can test a variety of player skills, including accuracy, logical reasoning, and creative problem solving

Abbreviations

PDE	Partial Differential Equation

3D three-dimensional

TGV Taylor Green Vortex

Declaration of Academic Achievement

The student will declare his/her research contribution and, as appropriate, those of colleagues or other contributors to the contents of the thesis.

~~Xinyu~~ developed most of the math and code for the optimization problem, Appendix A, B, C. I created the derivations in Appendix D, E, F

1 Introduction

1.1 Singularity Formation in Models of 3D Fluid Flows

The three-dimensional (3D) Navier-Stokes equations (NSE) are a set of partial differential equations (PDE) that describe the motion of homogeneous, viscous, incompressible fluids. The fluids described by the NSE can be modelled in domains with a wide variety of shape, dimension, and boundary conditions. Because of their ~~flexibility~~ they are utilized to model the behaviour of systems in fields as disparate as aero/hydrodynamics, weather prediction, biophysics, and astrophysics [4][6]. Despite the numerous fields that make use of the 3D Navier-Stokes equations, and extensive research examining their properties, ~~there is no known unique classical solution~~ [12]. In fact, this is the basis of the well known millennium problem from the Clay Institute which offers a ~~million dollar award~~ for the confirmation of the existence, or non-existence, of a solution for smooth initial conditions [13].

The Navier-Stokes system on the domain Ω is defined as

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} + \boldsymbol{\nabla} p - \nu \boldsymbol{\Delta} \boldsymbol{u} = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.1a}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{u} = 0 \qquad \text{in } \Omega \times [0, T], \tag{1.1b}$$

$$\boldsymbol{u}(x, 0) = \boldsymbol{\eta} \tag{1.1c}$$

where $\boldsymbol{u} = [u_1, u_2, u_3]^\intercal$ is the fluid velocity field, p is the scalar pressure, $\nu > 0$ is the coefficient of kinematic viscosity, and T is the ~~duration~~ of the time interval. The ~~value $\boldsymbol{\eta}$ is the initial divergence free velocity field of the fluid at time $t = 0$.~~ Finally, the $\boldsymbol{\nabla} = [\partial_1, \partial_2, \partial_3]^T$ operator is used to define the gradient of the pressure, the tensor $[\boldsymbol{\nabla}\boldsymbol{u}]_{ij} = \partial_j u_i$ for $i, j = 1, 2, 3$, and the divergence of the velocity field. It is common for scientific research that focuses the fundamental properties of the Navier-Stokes equations to utilize either the unbounded domain $\Omega := \mathbb{R}^3$, or the periodic 3D torus $\Omega := \mathbb{T}^3_L$ with $L > 0$ [12]. For this analysis, we use the unit 3D torus with periodic boundary conditions.

A central question in fluid mechanics is whether or not the Navier-Stokes equations are globally well-posed or if singularities can form from smooth initial conditions. A singularity occurs when the derivative of the velocity field becomes infinite at a single point [11]. The formation of a singularity in a fluid flow, from smooth initial conditions, would invalidate the Navier-Stokes equations as an accurate description of fluid flows [12]. The question of singularity formation from smooth initial conditions ~~has also not been answered~~ for the Euler equations [7]. The Euler equations are the inviscid form of the Navier-Stokes equations, obtained by setting $\nu = 0$

$$\partial_t \boldsymbol{u} + \boldsymbol{u} \cdot \boldsymbol{\nabla} \boldsymbol{u} + \boldsymbol{\nabla} p = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.2a}$$

$$\boldsymbol{\nabla} \cdot \mathbf{u} = 0 \qquad \text{in } \Omega \times [0, T], \tag{1.2b}$$

$$\boldsymbol{u}(x, 0) = \boldsymbol{\eta} \tag{1.2c}$$

The Euler are often studied because they are simpler to analyse than the Navier-Stokes equations. Additionally, there is numerical evidence that indicates singularity formation in Euler is possible from smooth initial conditions [2].

1.2 3D Euler-Voigt Equations

The Euler-Voigt equations are an inviscid regularization of the Euler equations (1.2) which have been shown to be globally well-posed and so do not form singularities from smooth initial conditions [3]. The Euler-Voigt equations are defined as

$$-\alpha^2 \partial_t \boldsymbol{\Delta} \boldsymbol{u}_\alpha + \partial_t \boldsymbol{u}_\alpha + \boldsymbol{u}_\alpha \cdot \boldsymbol{\nabla} \boldsymbol{u}_\alpha + \boldsymbol{\nabla} p_\alpha = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.3a}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{u}_\alpha = 0 \qquad \text{in } \Omega \times (0, T], \tag{1.3b}$$

$$\boldsymbol{u}_\alpha(x, 0) = \boldsymbol{\eta}_\alpha. \tag{1.3c}$$

~~The regularization of the Euler-Voigt equations~~ is accomplished with the addition of the term $-\alpha^2 \partial_t \boldsymbol{\Delta} \boldsymbol{u}_\alpha$ where $\alpha > 0$ is the regularization parameter with units of length. This term ~~is used to stabilize simulations without adding artificial viscosity,~~ which would remove energy from the system [8]. The regularization term acts as a spectral filter and suppresses the development of small scale structures with length-scale smaller than α. We introduce the subscript α to distinguish between solutions of (1.2) and (1.3).

For (1.2) the quantity $\|\boldsymbol{u}\|_{L^2}^2$ is conserved so long as the solution remains well-posed. For (1.3) the conserved quantity is known ~~as the α-energy.~~ The α-energy is defined as

$$E_\alpha(t) := \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0). \tag{1.4}$$

The goal of this analysis is to search for numerical evidence of singularity formation in the Euler equations from smooth initial conditions. This search will be performed with a PDE constrained optimization problem which will identify initial condition that produce 'extreme' behaviour in Euler-Voigt flows.

1.3 Finite Time Blow-up Criterion

In order to begin our search for finite-time singularity formation, we must first describe the ~~criterion~~ used to ~~identify~~ singularity formation in Euler flows. There are many different methods for determining if singularity formation has occurred, the most well known of which is the Beale-Kato-Majda (BKM) criterion [1] which states: A Smooth solution u of the Euler system develops a singularity at $t = t_0$ if and only if

$$\lim_{t \to t_0} \int_0^t \|\boldsymbol{\omega}\|_{L^\infty} d\tau = \infty, \qquad \boldsymbol{\omega} = \nabla \times \boldsymbol{u} \tag{1.5}$$

In [9] a new criterion for identifying singularity formation in Euler ~~equations~~ was developed which utilizes the Euler-Voigt equations. This criterion ~~began by proving~~

the convergence of the solutions of (1.3) to the solutions of (1.2) as $\alpha \to 0$.

Theorem 1.3.1. *Given initial data* $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ *for some* $s \geq 3$, *let* $\boldsymbol{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$ *be the corresponding solutions to (1.2) and (1.3), respectively, where* $0 < T < T_{max}$ *and* T_{max} *is the maximal time for which a solution to (1.2) exists and is unique. For all* $t \in [0,T]$, *we have*

$$\|\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \leq \left(\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2\right) e^{Ct} + C\alpha^2(e^{Ct} - 1). \tag{1.6}$$

Based on this theorem we know that if

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \text{as } \alpha \to 0, \tag{1.7}$$

then for all $t \in [0,T]$,

$$\|\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \to 0. \tag{1.8}$$

Therefore solutions of Euler-Voigt (1.3) converge to Euler (1.2) as $\alpha \to 0$. Using this ~~convergence [8] proved the following theorem.~~

Theorem 1.3.2. *Let* $\boldsymbol{\eta}_\alpha \in H^s$, $s \geq 3$ *with* $\boldsymbol{\nabla} \cdot \eta_\alpha = 0$ *and let* $\boldsymbol{u}_\alpha$ *be the corresponding unique solution of (1.3). Suppose that there exists a time* $T^* > 0$ *such that*

$$\limsup_{\alpha \to 0^+} \left(\alpha \sup_{t \in [0,T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2} \right) > 0. \tag{1.9}$$

Then the Euler equations (1.2) with the initial condition $\boldsymbol{\eta}_\alpha$ *must develop a singularity within the interval* $[0, T^*]$.

We note that this theorem relies on (1.2) and (1.3) starting from the same initial condition, i.e. $\boldsymbol{\eta} = \boldsymbol{\eta}_\alpha$. This necessarily satisfies (1.7) as $\alpha \to 0$. However, in order to use this ~~convergence for our optimization search we must generalize the singularity formation theorem to use (1.7)~~ where $\boldsymbol{\eta} \neq \boldsymbol{\eta}_\alpha$. Therefore, we ~~slightly~~ modify Theorem 5.2 from [9] as follows. Special thanks to Dr. Xinyu Zhao for developing this theorem and proof.

Theorem 1.3.3. *Given initial data* $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ *for some* $s \geq 3$ *satisfying*

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \text{as } \alpha \to 0, \tag{1.10}$$

let $\boldsymbol{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$ *be the corresponding solutions to (1.2) and (1.3), respectively. If there exists a finite time* $T^* > 0$ *such that the solutions* $\boldsymbol{u}_\alpha$ ~~*satisfies*~~

$$\limsup_{\alpha \to 0} \left(\alpha^2 \sup_{t \in [0,T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) > 0, \tag{1.11}$$

then $\boldsymbol{u}$ *develops a singularity in the interval* $[0, T^*]$.

Proof. Suppose that the solution $\boldsymbol{u}$ of the Euler equations (1.2) with the initial condition $\boldsymbol{\eta}$ is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have

$$\|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{\eta}_\alpha\|_{L^2}^2, \tag{1.12}$$

thus

$$\alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{\eta}_\alpha\|_{L^2}^2 - \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2. \tag{1.13}$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$\sup_{t\in[0,T^*]} \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2. \tag{1.14}$$

Taking the limsup as $\alpha \to 0$, we obtain ~~that~~

$$\limsup_{\alpha\to 0} \left(\sup_{t\in[0,T^*]} \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) = \limsup_{\alpha\to 0} \left(\|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right). \tag{1.15}$$

We now make use of Theorem (1.3.1), which demonstrates that the convergence of the solutions of the Euler-Voigt equations (1.3) to the solution of the Euler equations (1.2) is uniform on $[0, T^*]$. Therefore we can reduce the above equation to

$$\limsup_{\alpha\to 0} \left(\sup_{t\in[0,T^*]} \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) = \|\boldsymbol{\eta}\|_{L^2}^2 - \|\boldsymbol{u}(t)\|_{L^2}^2. \tag{1.16}$$

Since the energy of regular solutions of the Euler equations is conserved, we have

$$\limsup_{\alpha\to 0} \left(\sup_{t\in[0,T^*]} \alpha^2 \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) = 0, \tag{1.17}$$

which contradicts the hypothesis in the theorem that the above expression is positive. ~~The second~~ equality holds because

$$\begin{aligned}
\limsup_{\alpha\to 0} \left| \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2} - \|\boldsymbol{\eta}\|_{L^2} \right| &\le \limsup_{\alpha\to 0} \|\boldsymbol{u}_\alpha(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\
&\le \limsup_{\alpha\to 0} \|\boldsymbol{u}_\alpha(t_\alpha) - \boldsymbol{u}(t_\alpha)\|_{L^2}^2 + \|\boldsymbol{u}(t_\alpha) - \boldsymbol{\eta}\|_{L^2}^2 \\
&= 0.
\end{aligned}$$

□

1.4 Initial Conditions

We have established the criterion which we will use identify singularity formation in ~~fluid~~ flows from smooth initial conditions. In order to formulate our optimization problem we must now describe the function space we perform our search in. The theorems (1.3.1), (1.3.2), and (1.3.3) presented in Section (1.3) require ~~smooth~~ initial conditions $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$. However, the pseudo-spectral methods used in this study require the functions to be analytic in order to have exponential convergence. For the Navier-Stokes equations (1.1), the ~~viscosity~~ term acts to instantaneously ~~spatially smooth the initial condition. Therefore, a H^3 smooth initial condition will become Gevrey regular at the first time step and so can benefit from~~ the pseudo-spectral methods.

For the Euler equations (1.2) there is no viscosity and so the solution simply stays at the same level of smoothness as the initial condition. For Euler-Voigt the regularization term keeps the solution globally well-posed but does not ~~smooth the initial condition~~ [9]. Therefore, we must have analytic initial conditions to make use of the pseudo-spectral methods.

Finally, a requirement ~~of the Euler-Voigt equations~~ is that the fluid is divergence free at all times, including the initial conditions. Due to this, we assume without loss of generality that the initial conditions have a mean of zero. In order to identify initial conditions that satisfy these criteria, we first consider analytic initial conditions that belong to the Gevrey class

$$G^\sigma := \left\{ \boldsymbol{v} \in H^s(\mathbb{T}^3) : \|\boldsymbol{v}\|_{G^\sigma} = \langle \boldsymbol{v}, \boldsymbol{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0\cancel{.} \tag{1.19}$$

~~We select the following inner product, expressed in both summation and integral form,~~

$$\forall \boldsymbol{v}, \boldsymbol{u} \in G^\sigma, \quad \langle \boldsymbol{v}, \boldsymbol{u} \rangle_{G^\sigma} := \sum_{\boldsymbol{j} \in \mathbb{Z}^3} (1 + |2\pi \boldsymbol{j}|^2)^s e^{4\pi\sigma|\boldsymbol{j}|} \hat{\boldsymbol{v}}_{\boldsymbol{j}} \cdot \overline{\hat{\boldsymbol{u}}_{\boldsymbol{j}}} \tag{1.20a}$$

$$= \int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma|D|} \boldsymbol{v} \cdot \boldsymbol{u} \, d\boldsymbol{x}, \tag{1.20b}$$

where hat denotes the Fourier coefficient, overbar denotes complex conjugation, and the operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D|\boldsymbol{v}}\right]_{\boldsymbol{j}} := 2\pi|\boldsymbol{j}|\hat{\boldsymbol{v}}_{\boldsymbol{j}}, \qquad \left[\widehat{e^{\sigma|D|}\boldsymbol{v}}\right]_{\boldsymbol{j}} := e^{2\pi|\boldsymbol{j}|}\hat{\boldsymbol{v}}_{\boldsymbol{j}}. \tag{1.21}$$

The parameter σ determines the ~~size~~ of the Gevrey space

$$G^{\sigma_2} \subset G^{\sigma_1} \subset G^0 = H^s, \quad 0 < \sigma_1 < \sigma_2, \tag{1.22}$$

which is proved in Appendix (D). The value of σ is chosen in Section (3.5). In order to satisfy the divergence free condition, we introduce the following subspace in which optimal initial conditions will be sought

$$V := \{\boldsymbol{v} \in G^\sigma : \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0, \quad \int_{\mathbb{T}^3} \boldsymbol{v} \, d\boldsymbol{x} = 0\}. \tag{1.23}$$

Finally, we note that both (1.2) and (1.3) possess the scaling property, given a solution $\boldsymbol{u}_{(\alpha)}$ we can construct a new solution

$$\tilde{\boldsymbol{u}}(\boldsymbol{y},\tau) = \frac{1}{\gamma}\boldsymbol{u}(\beta\boldsymbol{x},\lambda t), \tag{1.24}$$

which is a valid solution of the Euler-Voigt equations (1.3) when $\lambda,\gamma,\beta > 0$ satisfy $\beta/\lambda\gamma = 1$. Our goal is to find an time interval [0,T] such that singularity formation ~~occurs~~ in that window. Due to this scaling property of the Euler-Voigt equations (1.24), the blow-up time of the solution will depend on the 'size' of our solution (γ) and the spatial ~~scaling~~ (β). Our domain size will not change, and so in order to keep consistent time scaling (λ) we must fix γ between all initial conditions. To accomplish this, we restrict our discussion to initial conditions with $\dot{H}^1$ seminorm which belong to a closed manifold $\cancel{\mathcal{M}_1} \subset V$ defined as

$$\cancel{\mathcal{M}_1} = \{\boldsymbol{v} \in V : \|\boldsymbol{v}\|_{\dot{H}^1} = C\}, \tag{1.25}$$

where C is a constant. The question we must answer now is what value of C ~~do~~ we use. To determine this we look to our first initial condition, the 3D Taylor Green Vortex (TGV). The TGV in the domain $x \in [0, L]^3$ has the form

$$\boldsymbol{\eta}_{TGV}(x) = \begin{bmatrix} V_0 \sin(2\pi x/L)\cos(2\pi y/L)\cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L)\sin(2\pi y/L)\cos(2\pi z/L) \\ 0\cancel{.} \end{bmatrix} \tag{1.26}$$

The $\dot{H}^1$-seminorm of (1.26) is

$$\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \sqrt{3L}\pi V_0, \tag{1.27}$$

which is proven in Appendix (E). In addition, the BKM criterion uses the $\|\boldsymbol{\omega}\|_{L^\infty}$ to determine singularity formation. For the TGV

$$\|\boldsymbol{\omega}\|_{L^\infty} = \|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = 4\pi V_0/L, \tag{1.28}$$

which is proven in Appendix (F). In [2], the authors ~~demonstrated~~ numerical evidence that blow-up of (1.2) ~~occurs at~~ $T^* \approx 4.4$ for initial condition (1.26) with parameters $L = 1$ and $V_0 = 1$. ~~Our goal is~~ to match ~~their~~ time window so that singularity formation identified by (1.3.3) occurs at approximately the same time as [2]. Therefore, we set $\lambda = 1$. Our ~~numerical models are performed~~ on the domain $\cancel{x} \in [0,1]^3$ and so we set $\beta = 1/2\pi$. Therefore, from (1.24) we set $\gamma = 1/2\pi$. The model we are aiming to match uses $V_0 = 1$ and so to match the time window we use $V_0 = 1/2\pi$.

These ~~parameters give our model an initial seminorm~~ $\|\eta_{TGV}\|_{\dot{H}^1} = \sqrt{3}/2$ and initial vorticity $\|\nabla \times \eta_{TGV}\|_{L^\infty} = 2$. Therefore, we can restrict our discussion of the TGV to initial conditions with $\dot{H}^1$ seminorm which belong to a closed manifold $\cancel{\mathcal{M}_1 \in V}$ ~~defined as~~

$$\cancel{\mathcal{M}_1 = \{\boldsymbol{v} \in V : \|\boldsymbol{v}\|_{\dot{H}^1} = \sqrt{3}/2\}.} \tag{1.29}$$

In order to select initial conditions within this manifold, we will first select initial conditions $\boldsymbol{\eta} \in V$ and then apply a retraction function to ensure it has $\|\cdot\|_{\dot{H}^1} = \sqrt{3}/2$. To verify that this selection for the initial $\dot{H}^1$ seminorm is correct we compare this

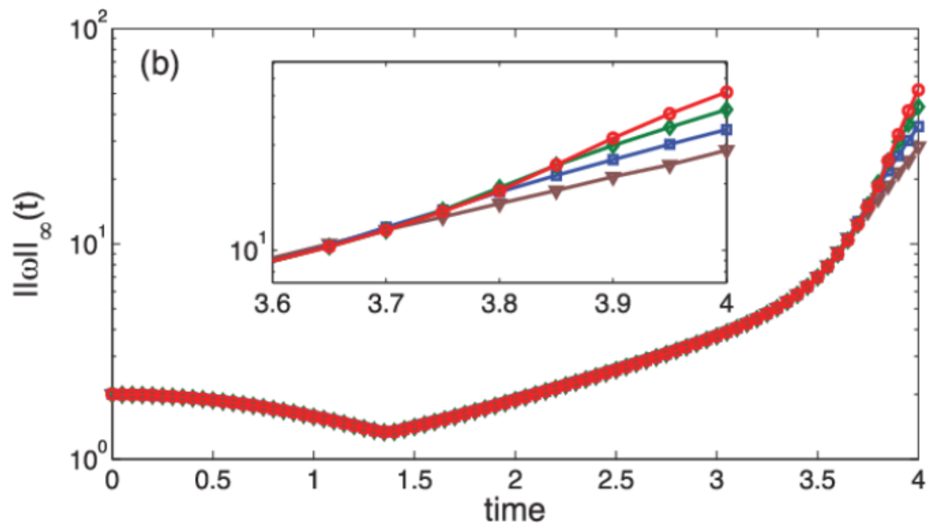

Figure 1.1: (Bustamente et al. 2012 Fig. 1b) Maximum of vorticity $\|\omega(\cdot,t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 512^3 (brown triangles), 1024^3 (blue squares), 2048^3 (green diamonds), and 4096^3 (red circles).

scaling to the results of [2]. Figure 1b of [2] shows the evolution of the $L-\infty$ norm of the vorticity over time (??). We present our reproduction in (??).

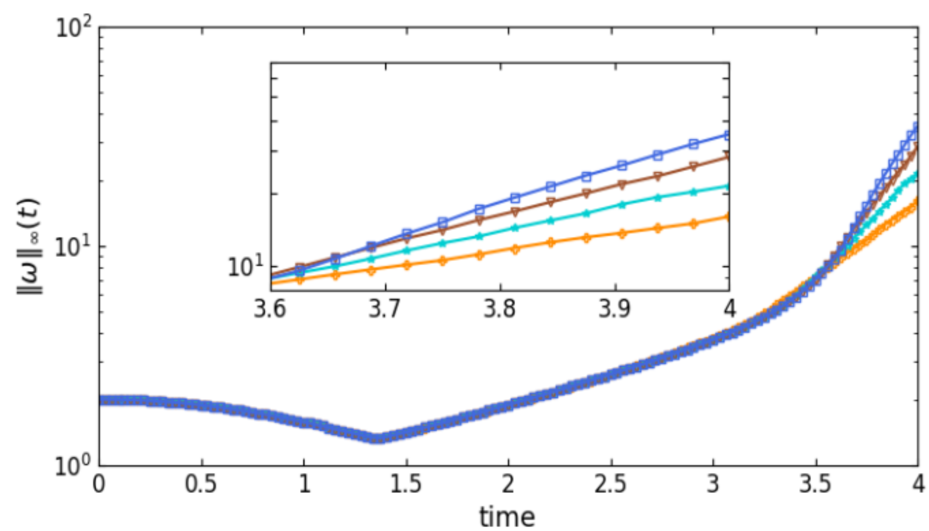

Figure 1.2: Maximum of vorticity $\|\omega(\cdot,t)\|_\infty$. Results from runs performed at different resolutions are displayed together: 128^3 (orange plus), 256^3 (cyan star), 512^3 (brown triangles), 1024^3 (blue squares).

These figures show that the evolution of the vorticity in our scaled model match exactly to the results of ([2]). Therefore, our selection for the manifold M_1 is correct.

1.5 Outline

In Section 2 we present the formulation of the optimization problem. This includes the description of our objective functional, the gradient calculation, and the Riemannian conjugate gradient method used to compute our search direction. In Section 3 we describe the numerical methods used to perform the calculations, including spatial resolution, time step size, Gevrey parameters and the regularization parameter. In addition we validate the model using the spectrum, kappa test, and conservation of $\alpha-$Energy. In Section 4 we present our results. We begin with the pre-optimization results and then present our optimization results as $\alpha \to 0$. Finally, in section 4 we summarize our results.

2 Statement of the Optimization Problem

2.1 Formulation of the Problem

PDE-constrained optimization problem. In our case the PDE constraint is the Euler-Voigt equations (1.3).

With our initial conditions and blowup criteria defined, we now define the objection functional $\Phi_{\alpha,T} : V \to \mathbb{R}^+$ as

$$\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) := \|\nabla \boldsymbol{u}_\alpha(T; \boldsymbol{\eta}_\alpha)\|_{L^2}^2, \tag{2.30a}$$

$$:= \|\boldsymbol{u}_\alpha(T; \boldsymbol{\eta}_\alpha)\|_{\dot{H}^1}^2, \tag{2.30b}$$

where $\boldsymbol{u}_\alpha(T; \boldsymbol{\eta}_\alpha)$ is the solution of the Euler-Voigt equations obtained at time T with the initial condition $\boldsymbol{\eta}_\alpha \in V$ and regularization parameter α. We can now formulate our optimization problem. Given $\alpha, T \in \mathbb{R}^+$, find

$$\tilde{\boldsymbol{\eta}}_{\alpha,T} = \underset{\boldsymbol{\eta}_\alpha \in M_1}{\operatorname{argmax}} \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha). \tag{2.31}$$

The objective functional is maximized using an Riemannian conjugate gradient iterative relation described in Section (2.3).

"A manifold whose tangent spaces are endowed with a smoothly varying inner product is called a Riemannian manifold (absil 2008)"

The same method was used in (XINYU PAPER) to construct a PDE-constrained optimization problem for the Euler equations to search for initial conditions $\boldsymbol{\eta}$ with unit $\dot{H}^3$ seminorm such that the corresponding optimal solution achieves a maximum $\dot{H}^3$ seminorm at a prescribed time Ts.

accelerates the gradient ascent method which is necessary given the computational cost of each iteration.

The solution to the problem is the value of $\boldsymbol{\eta}^n_{\alpha,T}$ as the iteration number $n \to \infty$. Our goal is to first fix T and solve the optimization problem for a sequence of decrease values of α that converge to 0. We examine the results for a value of T before the expected point singularity formation should occur and again for a T after. From Theorem (1.3.3) if

$$\|\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T}\|^2_{L^2} + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \tilde{\boldsymbol{\eta}}_{\alpha,T})\|^2_{L^2} \to 0, \quad \text{as } \alpha \to 0, \tag{2.32}$$

and

$$\limsup_{\alpha \to 0} \left(\alpha^2 \sup_{t \in [0,T^*]} \Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T}) \right) > 0, \tag{2.33}$$

then $\boldsymbol{u}$ develops a singularity in the interval $[0, T^*]$. FROM LITERATURE expected time of singularity formation in Euler is $T^* \approx 4$. Therefore, we perform the optimization problem for $T = 3$ and $T = 5$. We expect that upon optimization the results of

$T = 3$ should indicate blow-up will not occur and for $T = 5$ it will occur.

2.2 Gradient of Objective Function

The optimization problem is solved using an iterative relation which uses an optimal search direction and distance to create the next iteration's initial condition. We compute the optimal search direction using the gradient of the objection functional $\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)$. This value is computed using an adjoint method which depends on solving a properly defined adjoint system backwards in time. We first introduce the Gâteaux (directional) differential $\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}_\alpha{}') : V \times V \to \mathbb{R}$. The first order finite difference equation for the Gateaux differential is,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}_\alpha{}') := \lim_{\epsilon\to 0}\frac{1}{\epsilon}\left[\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon\boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha)\right]. \tag{2.34}$$

This represents the change in $\Phi_{\alpha,T}$ from a small perturbation proportional to $\boldsymbol{\eta}'_\alpha$ applied to the initial condition $\boldsymbol{\eta}_\alpha$. By fixing the initial condition $\boldsymbol{\eta}_\alpha$, we can view the Gâteaux differential as a bounded linear functional on V (XINYU PAPER). Therefore, it can also be represented as an inner product on the Gevrey space using the Riesz representation theorem **(author?)** [10]

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}_\alpha{}') = \langle\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha),\boldsymbol{\eta}_\alpha{}'\rangle_{G^\sigma}. \tag{2.35}$$

Substituting the second definition of the objective functional (2.30) into the finite difference form of the Gâteaux differential, we derive the Gâteaux differential in the form of the $\dot{H}^1$ inner product and the L^2 inner product,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\cdot,T), \boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1} = \left\langle \boldsymbol{u}^*_\alpha(\cdot,0), \boldsymbol{u}'_\alpha(\cdot,0)\right\rangle_{L^2} \tag{2.36}$$

The complete derivations for both forms are shown in Appendix (A). In (2.36) the value $\boldsymbol{u}'_\alpha$ is the solution of the linearization of the Euler-Voigt system (1.3) around its solution $\boldsymbol{u}_\alpha$. The linearization of the Euler-Voigt system is defined as

$$\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix} := \begin{bmatrix}\partial_t \boldsymbol{u}'_\alpha + (I-\alpha^2\Delta)^{-1}(\boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p') = 0\\ \nabla\cdot\boldsymbol{u}'_\alpha = 0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \tag{2.37a}$$

$$\boldsymbol{u}'_\alpha(x,0) = {\boldsymbol{\eta}_\alpha}' \tag{2.37b}$$

where ${\boldsymbol{\eta}_\alpha}'$ and p' are the perturbations of the initial condition and pressure, respectively. The derivation of the linearized form of the Euler-Voigt equations is shown in Appendix (B). The value $\boldsymbol{u}^*_\alpha$ in (2.36) is the solution of the adjoint system

$$\mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*_\alpha\end{bmatrix} := \begin{bmatrix}-\partial_t \boldsymbol{u}^*_\alpha - (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha + (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha)^\intercal)\boldsymbol{u}_\alpha - \boldsymbol{\nabla}p^*_\alpha\\ -\boldsymbol{\nabla}\cdot\boldsymbol{u}^*_\alpha\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix} \tag{2.38a}$$

$$\boldsymbol{u}^*_\alpha(\boldsymbol{x},T) = 2|D|^2\boldsymbol{u}_\alpha(\boldsymbol{x},T) \tag{2.38b}$$

which is derived in Appendix (C) by performing integration by parts, in both space and time, on the following duality-pairing relation

$$\begin{aligned}\left(\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix},\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right) &:= \int_0^T\int_{\mathbb{T}^3}\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix},\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\,d\boldsymbol{x}\,dt\\ &= \left(\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix},\mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right)+\int_{\mathbb{T}^3}\boldsymbol{u}'_\alpha(\boldsymbol{x},T)\cdot\boldsymbol{u}^*_\alpha(\boldsymbol{x},T)\,d\boldsymbol{x}\\ &\quad-\int_{\mathbb{T}^3}\boldsymbol{\eta}'(\boldsymbol{x})\cdot\boldsymbol{u}^*_\alpha(\boldsymbol{x},0)\,d\boldsymbol{x}\\ &=0.\end{aligned} \tag{2.39}$$

Using the adjoint system we can now define the gradient. We take the Gâteaux differential in the form of the L^2 inner product (2.36) and in the Riesz representation form (2.35)

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha;\boldsymbol{\eta}_\alpha{}') = 2\left\langle\boldsymbol{u}^*_\alpha(\cdot,0),\boldsymbol{u}'_\alpha(\cdot,0)\right\rangle_{L^2} = \left\langle\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha),\boldsymbol{\eta}_\alpha{}'\right\rangle_{G^\sigma}. \tag{2.40}$$

Expressing the L^2 inner product in its integral form,

$$\left\langle\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}),\boldsymbol{\eta}'\right\rangle_{G^\sigma} = \int_{\mathbb{T}^3}\boldsymbol{u}^*_\alpha(x,0)\cdot\boldsymbol{\eta}'(x)\,dx \tag{2.41a}$$

$$= \int_{\mathbb{T}^3}(1+|D|^2)^3e^{2\sigma|D|}\left[(1+|D|^2)^{-3}e^{-2\sigma|D|}\boldsymbol{u}^*_\alpha(x,0)\right]\cdot\boldsymbol{\eta}'(x)\,dx \tag{2.41b}$$

$$= \left\langle(1+|D|^2)^{-3}e^{-2\sigma|D|}\boldsymbol{u}^*_\alpha(x,0),\boldsymbol{\eta}'(x)\right\rangle_{G^\sigma} \tag{2.41c}$$

This gives us the Gâteaux differential in the form of the Gevrey inner product

$$\left\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}), \boldsymbol{\eta}' \right\rangle_{G^\sigma} = \left\langle (1+|D|^2)^{-3} e^{-2\sigma|D|} \boldsymbol{u}^*_\alpha(x,0), \boldsymbol{\eta}'(x) \right\rangle_{G^\sigma}. \tag{2.42}$$

Therefore, the gradient of the objective functional is

$$\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) = (1+|D|^2)^{-3} e^{-2\sigma|D|} \boldsymbol{u}^*_\alpha(x,0) \tag{2.43}$$

The adjoint system (2.38) is a terminal value problem. Therefore, to compute the gradient we first evolve the Euler-Voigt system forward from $\boldsymbol{\eta}_\alpha$ to compute $\boldsymbol{u}_\alpha(\boldsymbol{x}, T)$, from which we obtain the terminal value of (2.38). Finally, we evolve the adjoint system backwards in time to obtain the initial value $\boldsymbol{u}^*_\alpha(\boldsymbol{x}, 0)$ which is used to compute (2.43).

2.3 Riemannian Conjugate Gradient Method

The optimization solution looks for an initial condition $\boldsymbol{\eta}_\alpha \in M_1$ that maximizes our objective functional. Therefore, not only must our first initial condition $\boldsymbol{\eta}^0_{\alpha,T}$ have a unit $\dot{H}^1$ seminorm, but so to must each following iteration $\boldsymbol{\eta}^{n+1}_{\alpha,T}$. We ensure this constraint is upheld using a Riemannian conjugate gradient method modified from the method used in [12] and [5]. This method has three steps:

1. Project the gradient $\nabla\Phi_T(\boldsymbol{\eta}_\alpha)$ onto the tangent space to $\mathcal{M}_1$ at $\boldsymbol{\eta}_\alpha$
2. Perform a suitable vector transport operation in order to construct a Riemannian conjugate ascent direction using the previous search direction
3. Retract the resulting state back to the constraint manifold $\mathcal{M}_1$

We first define the tangent space to $\mathcal{M}_1$ as

$$\mathcal{T}_{\boldsymbol{v}}\mathcal{M}_1 = \left\{\boldsymbol{z} \in V : \langle \boldsymbol{v}, \boldsymbol{z}\rangle_{\dot{H}^1} = 0\right\}.$$

. For any $\boldsymbol{z} \in \mathcal{T}_{\boldsymbol{v}}\mathcal{M}_1$, this condition can be expressed as

$$\langle \boldsymbol{v}, \boldsymbol{z}\rangle_{\dot{H}^1} = \int_{\mathbb{T}^3} |D|\boldsymbol{v} \cdot |D|\boldsymbol{z} d\boldsymbol{x} \tag{2.44a}$$

$$= \int_{\mathbb{T}^3} (1+|D|^2)^3 e^{2\sigma|D|} \left((1+|D|^2)^{-3} e^{-2\sigma|D|}|D|^2\boldsymbol{v}\right) \cdot \boldsymbol{z}\, d\boldsymbol{x} \tag{2.44b}$$

$$= \left\langle (1+|D|^2)^{-3} e^{-2\sigma|D|}|D|^2\boldsymbol{v}, \boldsymbol{z}\right\rangle_{G^\sigma} = 0. \tag{2.44c}$$

Thus, at each point $\boldsymbol{v} \in M_1$, the tangent space $\mathcal{T}_{\boldsymbol{v}}, \mathcal{M}_1$ can be characterized using the unit 'normal' vector given by

$$\boldsymbol{n}_{\boldsymbol{v}} = \frac{(1+|D|^2)^{-3} e^{-2\sigma|D|}|D|^2\boldsymbol{v}}{\|(1+|D|^2)^{-3} e^{-2\sigma|D|}|D|^2\boldsymbol{v}\|_{G^\sigma}} \tag{2.45}$$

which allows us to construct the projection operator $\mathcal{P}_{\mathcal{T}_{\boldsymbol{v}}M_1} : V \to \mathcal{T}_{\boldsymbol{v}}\mathcal{M}_1$,

$$\forall \boldsymbol{w} \in V, \quad \mathcal{P}_{\mathcal{T}_{\boldsymbol{v}}\mathcal{M}_1}\boldsymbol{w} = \boldsymbol{w} - \langle \boldsymbol{n}_{\boldsymbol{v}}, \boldsymbol{w}\rangle_{G^\sigma} \boldsymbol{n}_{\boldsymbol{v}}. \tag{2.46}$$

As stated previously, the initial conditions at each iteration must have a unit $\dot{H}^1$. In order to ensure this we introduce the Retraction operator

$$\mathbf{R}(\boldsymbol{v}) = \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|_{\dot{H}^1}}, \qquad \boldsymbol{v} \neq \mathbf{0}. \tag{2.47}$$

The Reimannian method uses the search direction of the n-th iteration to compute the search direction of the $(n+1)$-th iteration. This is done by a linear combination of the current direction and a 'momentum' term based on the previous iteration's search direction. However, the two search directions belong to different tangent spaces to $\mathcal{M}_1$ and thus cannot be directly added. In order to allow algebraic operations to be performed on the vectors of the two tangent spaces we introduce the vector transport operation

$$\mathcal{TM}_1 \oplus \mathcal{TM}_1 \to \mathcal{TM}_1 : (\boldsymbol{\varphi}, \boldsymbol{\xi}) \mapsto \Gamma_{\boldsymbol{\varphi}}(\boldsymbol{\xi}) \in \mathcal{TM}_1,$$

where $\mathcal{TM}_1 = \bigcup_{\boldsymbol{v} \in \mathcal{M}_1} \mathcal{T}_v\mathcal{M}_1$ is the tangent bundle on $\mathcal{M}_1$. The operation transports the vector field $\boldsymbol{\xi}$ along the manifold M_1 in the direction of $\boldsymbol{\varphi}$ [CITATION]. This allows us to map search directions between the tangent spaces of two consecutive iterations.

3 Numerical Methods

3.1 Spatial Discretization & Spectral Methods

Simulations were computed using a standard pseudospectral fourier method on a periodic unit cube, discretized in a uniform grid of spatial resolutions $N = 128^3, 256^3, 512^3, 1024^3$. Derivatives are evaluated in the Fourier space and nonlinear products are computed in the physical space. The standard 2/3 dealiasing rule is also applied. Our interest is in the limit $\alpha \to 0$. Therefore, we use the regularization parameters $\alpha = 2^{-6}, 2^{-7}, 2^{-8}, 2^{-9}, 2^{-10}$.

3.2 Time Stepping

Time stepping is performed with the explicit fourth-order Runge-Kutta (RK4) method. The determination of our time step size Δt is based on three factors. First, we consider the numerical stability of the models. Second, we consider the accuracy of the solution, determined by the relative error of the objective functional (2.30) to the results for a minimum step size. Finally, we consider the conservation of the α-energy, determined by the maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$. The RK4 method does not conserve energy, and so the α-energy of our model gradually decreases over time. These factors are analysed for each resolution and value of α for the Taylor Green

Vortex.

The goal of our optimization problem is to identify initial conditions $\tilde{\boldsymbol{\eta}}_{\alpha,T}$ that produce 'extreme' behaviour in solutions to the Euler-Voigt equations. Therefore, the Taylor Green Vortex will by definition be the least extreme initial condition examined. To ensure stability, accuracy, and conservation of α-energy are maintained upon optimization we must select a step size smaller than the minimum step size identified by these criteria.

3.2.1 Numerical Stability & Accuracy

We begin with the stability and accuracy of the Euler-Voigt model. To determine the stability and accuracy of the model at each step size we start from the iteration zero initial conditions and evaluate the objective functional (2.30) for the time window $T = 5$. First we do this for $\Delta t = 2^{-8}$ and designate the result as our 'correct' value Φ_c. We then compute the relative error of the objective functional for each value of Δt (Φ_{dt}) to Φ_c

$$\epsilon_{rel} = \left| \frac{\Phi_c - \Phi_{dt}}{\Phi_c} \right| . \tag{3.48}$$

For models that experienced numerical instability in the time window $[0, T]$ we set ϵ_{rel} to infinity.

Tables (3.1) to (3.4) show the results for ϵ_{rel} at the four resolutions utilized in this analysis. The tables shows that the accuracy of the objective functional decreases as $\alpha \to 0$ and increases as $\Delta t \to 0$. In addition, the accuracy of the model overall increases as the resolution increases. Finally, the size of the instability region does

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.0045	7.00E-05
2^{-3}	Inf	Inf	0.0572	0.0011	3.64E-06
2^{-4}	0.1662	0.1157	0.0332	4.85E-04	2.02E-07
2^{-5}	0.1093	0.0715	0.0181	2.33E-04	1.21E-08
2^{-6}	0.0593	0.0369	0.0085	1.02E-04	8.63E-10
2^{-7}	0.0228	0.0137	0.003	3.45E-05	9.44E-11
2^{-8}	0	0	0	0	0

Table 3.1: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 128^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.003	7.00E-05
2^{-3}	Inf	Inf	0.0112	1.82E-04	3.64E-06
2^{-4}	Inf	0.0652	0.0021	9.46E-06	2.01E-07
2^{-5}	0.1097	0.0348	7.13E-04	5.12E-07	1.17E-08
2^{-6}	0.0619	0.0171	3.08E-04	3.00E-08	7.01E-10
2^{-7}	0.025	0.0157	1.05E-04	1.96E-09	4.04E-11
2^{-8}	0	0	0	0	0

Table 3.2: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 256^3$ and time window $T = 5$.

increase with resolution, but only for the smallest value of α. For all other values of α, the stability is unaffected over the time window $t \in [0, T]$.

3.2.2 Conservation of α-Energy

The conservation of α-energy is also used to verify the model for a given resolution, regularization parameter, and time window. The tables below show the maximum

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.003	7.00E-05
2^{-3}	Inf	Inf	0.009	1.82E-04	3.64E-06
2^{-4}	Inf	0.0174	7.61E-04	9.45E-06	2.01E-07
2^{-5}	Inf	0.0036	3.63E-05	5.10E-07	1.17E-08
2^{-6}	0.0306	7.85E-04	1.72E-06	2.90E-08	7.01E-10
2^{-7}	0.0115	2.42E-04	8.42E-08	1.63E-09	4.04E-11
2^{-8}	0	0	0	0	0

Table 3.3: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 512^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	0.003	7.00E-05
2^{-3}	Inf	Inf	0.009	1.82E-04	3.64E-06
2^{-4}	Inf	0.015	7.61E-04	9.45E-06	2.01E-07
2^{-5}	Inf	0.0022	3.64E-05	5.10E-07	1.17E-08
2^{-6}	0.0039	2.64E-04	1.71E-06	2.90E-08	7.01E-10
2^{-7}	5.17E-04	4.92E-06	8.36E-08	1.63E-09	4.04E-11
2^{-8}	0	0	0	0	0

Table 3.4: Relative error of Φ_{dt} for each step size to Φ_c. Resolution $N = 1024^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	3.82E-04	8.73E-06
2^{-3}	Inf	Inf	1.94E-03	6.91E-05	3.72E-07
2^{-4}	3.38E-03	2.74E-03	1.11E-03	3.05E-05	1.76E-08
2^{-5}	2.38E-03	1.81E-03	6.49E-04	1.57E-05	9.59E-10
2^{-6}	1.51E-03	1.09E-03	3.56E-04	8.05E-06	7.55E-11
2^{-7}	8.67E-04	6.06E-04	1.87E-04	4.08E-06	1.47E-11
2^{-8}	4.69E-04	3.21E-04	9.59E-05	2.05E-06	6.00E-12

Table 3.5: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 128^3$ and time window $T = 5$.

Δt \ α	1/1024	2/1024	4/1024	8/1024	16/1024
2^{-2}	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	4.34E-04	1.49E-05	3.72E-07
2^{-4}	Inf	8.36E-04	5.61E-05	6.85E-07	1.75E-08
2^{-5}	9.63E-04	4.35E-04	1.80E-05	3.40E-08	9.09E-10
2^{-6}	6.33E-04	2.48E-04	8.97E-06	1.89E-09	4.95E-11
2^{-7}	3.83E-04	1.37E-04	4.59E-06	1.38E-10	2.89E-12
2^{-8}	2.14E-04	7.23E-05	2.33E-06	2.37E-11	2.95E-12

Table 3.6: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 256^3$ and time window $T = 5$.

relative error of the α-energy for each α and Δt,

$$\epsilon_\alpha = \max_{t\in[0,T]} \left| \frac{E_\alpha(t) - E_\alpha(0)}{E_\alpha(t)} \right|. \tag{3.49}$$

For models that experienced numerical instability in the time window $[0, T]$ we set ϵ_α to infinity.

The Tables show that the conservation of α-energy error ϵ_α increases as $\alpha \to 0$ and decreases as $\Delta t \to 0$. In addition, the error decreases as the resolution increases.

	α: 1/1024	2/1024	4/1024	8/1024	16/1024
$\Delta t = 2^{-2}$	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	3.85E-04	1.49E-05	3.72E-07
2^{-4}	Inf	2.72E-04	2.58E-05	6.85E-07	1.75E-08
2^{-5}	Inf	2.89E-05	1.12E-06	3.38E-08	9.09E-10
2^{-6}	1.58E-04	8.18E-06	5.00E-08	1.82E-09	4.95E-11
2^{-7}	9.02E-05	3.74E-06	2.47E-09	1.03E-10	3.44E-12
2^{-8}	4.98E-05	1.91E-06	1.42E-10	6.31E-12	2.69E-12

Table 3.7: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 512^3$ and time window $T = 5$.

	α: 1/1024	2/1024	4/1024	8/1024	16/1024
$\Delta t = 2^{-2}$	Inf	Inf	Inf	2.91E-04	8.73E-06
2^{-3}	Inf	Inf	3.85E-04	1.49E-05	3.72E-07
2^{-4}	Inf	2.53E-04	2.58E-05	6.85E-07	1.75E-08
2^{-5}	Inf	2.67E-05	1.12E-06	3.38E-08	9.09E-10
2^{-6}	1.46E-05	1.26E-06	4.99E-08	1.82E-09	4.95E-11
2^{-7}	2.53E-06	5.12E-08	2.45E-09	1.03E-10	3.25E-12
2^{-8}	1.06E-06	2.35E-09	1.30E-10	6.31E-12	2.78E-12

Table 3.8: Maximum relative error of $E_\alpha(t)$ to $E_\alpha(0)$ for each step size. Resolution $N = 1024^3$ and time window $T = 5$.

		α				
		1/1024	2/1024	4/1024	8/1024	16/1024
N	128^3	2^{-8}	2^{-8}	2^{-6}	2^{-5}	2^{-3}
	256^3	2^{-8}	2^{-8}	2^{-6}	2^{-4}	2^{-3}
	512^3	2^{-8}	2^{-7}	2^{-5}	2^{-4}	2^{-3}
	1024^3	2^{-8}	2^{-7}	2^{-5}	2^{-4}	2^{-3}

Table 3.9: Selection of Δt for each resolution and α.

Based on the results of the stability, accuracy, and α-energy conservation we can now select the appropriate time step size for each α and resolution, shown in Table (3.9). These step sizes are selected because they are the largest step sizes that ensure the stability, accuracy and α-energy conservation of the model from the Taylor Green Vortex.

3.3 Spectrum Validation

The energy spectrum of the solution $\boldsymbol{u}_\alpha(\boldsymbol{x}, t)$ is defined as,

$$e(k,t) := \frac{1}{2} \sum_{j \le |\boldsymbol{j}| < j+1} |\hat{\boldsymbol{u}}_{\alpha \boldsymbol{j}}(t)|^2, \quad k = 2\pi j, \quad j \in \mathbb{N}. \tag{3.50}$$

The regularization term of (1.3) acts as a filter which suppresses the growth of small-scale structures of length-scale smaller than the regularization parameter α. Therefore, the spectrum (3.50) is suppressed for wavenumbers $k > k_\alpha$ where

$$k_\alpha := \frac{2\pi}{\alpha}. \tag{3.51}$$

Due to this filter, the solution to (1.3) can remain well resolved. This is ideal for our analysis because it ensures that the model fully captures the small scale structures of the fluid. When this occurs we can be certain that our objective functional (2.30) is accurate. We begin by examining the spectrum of the solution from the Taylor Green Vortex on the time window $[0, T]$ for $T = 5$.

The plots () to () show the semi-log energy vs wavenumber graph of the solution over time.

initial condition shown below demonstrates the exponential decay expected by a analytic function. As the solution evolves in time the rate of decay decreases due to aliasing. While this can be reduced by anti-aliasing techniques, it cannot be entirely prevented

Low resolution and small alpha simulations are especially vulnerable to aliasing errors and the solutions stop being analytic before the chosen timescale.

The chosen timescales, alphas, and resolutions are validated in the graphs below which show that the spectral decay remains exponential and thus the solutions remain analytic.

3.4 Kappa Test Validation

The simulations are discretized in space using a psuedospectral method and time using a fourth order Runge Kutta method. The kappa test is a means of validating these discretization methods based on the validation method of [**?**]. We define the value κ as

$$\kappa(\epsilon) = \frac{\epsilon^{-1}\left[\Phi_{\alpha,T}(\boldsymbol{\eta} + \epsilon\boldsymbol{\eta}') - \Phi_{\alpha,T}(\boldsymbol{\eta})\right]}{\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}), \boldsymbol{\eta}' \rangle} \tag{3.52}$$

where $\Phi_{\alpha,T}$ is the objective functional (2.30), $\boldsymbol{\eta}$ is the initial condition, and $\nabla\Phi_{\alpha,T}$ is the gradient of the objective functional (2.43). The numerator is the first order finite difference approximation of the Gateaux differential. The denominator is the Reisz form of the differential that utilizes the adjoint system (2.38). If the numerical approximation methods are accurate, then we expect $\kappa \approx 1$.

A log-log plot of $|\kappa - 1|$ vs ϵ can be broken up into three regions, each dominated by different error sources. These are the discretization errors, round off errors, and truncation errors. The round off errors are caused by finite precision numerical representation and so are approximately of order $O(\epsilon^{-1})$. Therefore, we expect the value of κ to be dominated by round off errors for small ϵ values. The truncation errors are caused by using a first order finite difference formula to calculate the derivative. The truncation errors are of order $O(\epsilon)$. Therefore, we expect the value of κ to be dominated by the truncation errors for large values of ϵ. Between these two regions the value of κ will be determined by the discretization errors in the numerical solutions, which are themselves a product of the spatial and temporal resolution. With smaller values of Δx and Δt causing the error to decrease. This will cause the value of $\log_{10}|kappa - 1|$ to decrease as the resolution increases. Additionally, an increase in the resolution will decrease the range of ϵ over which the discretization errors dominate. Therefore, we expect the plot of κ vs ϵ to be of the general form shown in the graph below.

The following κ test graphs were generated using a fixed spacial resolution of 128 and

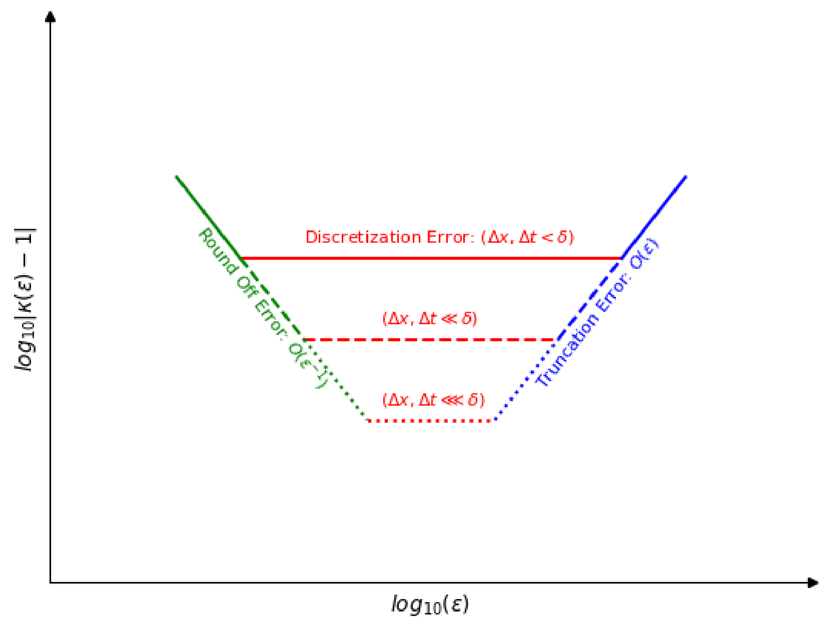

Figure 3.1: Plot showing the expected κ behaviour for different spatial and temporal resolution where δ represents a small arbitrary constant.

with varying temporal resolutions $\Delta t \in [2^{-4}, 2^{-12}]$. The initial conditions used are the 3D Taylor Green, 3D ABC, and 3D ABC Mixture, each with a random initial perturbation. The simulations were run for timescales of $T = 1$ and $T = 3$. The results are shown below.

The results from the 3D Taylor-Green κ test show the expected behaviour from the theoretical kappa test graph. The value of kappa increases at very small and large values of epsilon. The value of kappa approaches 1 as Δt decreases, and so the plateau region lowers and decreases in width. The T=1 time scale results follow very closely to the expected results while the T=3 time scale simulations show deviations from the the expected behaviour. These differences are due to the random initial direction of each simulation having more time to deviate from one another.

The kappa test results for the 3D ABC initial condition exhibit the same lowering of kappa as the time step decreases. However, unlike the Taylor Green initial condition, the results from the 3D ABC κ test have little to no plateau region. This indicates that the 3D ABC initial condition simulations are much more sensitive to the round off and truncation errors.

The results from the 3D ABC Mixture κ test show the same expected behaviour as the 3D Taylor Green kappa test.

The three kappa test results align well with the expected theoretical result. Therefore, we conclude that the discretization and time stepping techniques used in this analysis are valid and accurate.

3.5 Determination of Gevrey class size

In Section (1.4) we introduce the Gevrey space in which we search for initial conditions (1.19). The size of the space is determined by two variables, $s \geq 3$ and $\sigma > 0$. We use a fixed value of $s = 3$. In order to determine the value of σ we start with the optimization problem for $\alpha = 16/1024$, $T = 5$, and resolution $N = 128^3$. The problem is repeated for $\sigma =$ 1E-1, 1E-2, 1E-3, 1E-4, 1E-5.

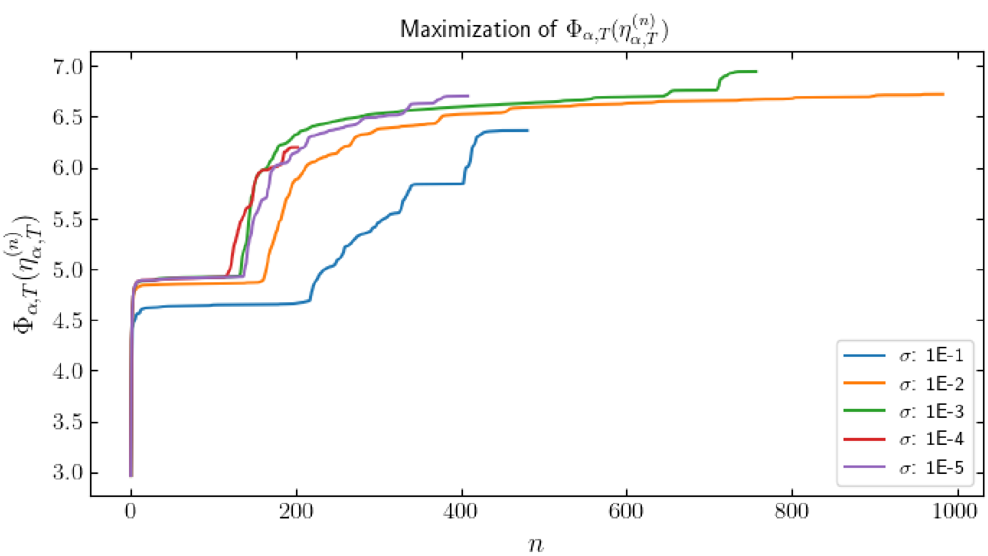

Figure 3.2: Maximization of $\Phi_{\alpha,T}(\boldsymbol{\eta}_{\alpha,T}^{(n)})$ vs iteration number. $\alpha = 16/1024$, $T = 5$, $N = 128^3$.

4 Results

The Theorem (XINYU THOREM) provide the criteria we use to confirm that singularity formation occurs in Euler as $\alpha \to 0$. BECAUSE OUR NUMERICAL MODELS MUST STOP AT SOME SMALL ALPHA OUR ALPHA SQUARED OBJECTIVE FUNCTION WILL NATURALLY BE POSITIVE. TO CONFIRM THAT OUR CRITERIA IS UPHELD IN THE LIMIT ALPHA GOING TO ZERO WE USE THE METHOD UTILIZED IN LARIOS ET AL 2018.

(COMPUTATIONAL INVESTIGATION) propose the following ansatz,

$$\sup_{t\in[0,T^*]} \|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2} \approx \mathcal{O}(\alpha^p), \tag{4.53}$$

for $T^* > 0$ sufficiently large and for some power p. From Theorem (XINYU THEOREM), if there exists a finite time $T^* > 0$ such that

$$\limsup_{\alpha\to 0} \left(\alpha^2 \sup_{t\in[0,T^*]} \|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) > 0. \tag{4.54}$$

then $\boldsymbol{u}$ develops a singularity in the interval $[0, T^*]$. From this ansatz, in order for singularity formation to occur, we must have $p \leq -1$.

We present numerical evidence that singularity formation does not occur from the Taylor Green Vortex as $\alpha \to 0$. In addition, we present numerical evidence that upon optimization ... RESULTS!

4.0.1 Pre-Optimization Results

We begin by presenting our pre-optimization results for the Taylor Green Vortex, i.e. $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$. FROM BRACHET AND BUSTAMANTE ET AL WE ANTICIPATE THAT BLOWUP SHOULD OCCUR AT $T^* \approx 4$. We compare the results at $T = 3$ and $T = 5$. Therefore, we expect that we should obtain $p > -1$ for $T = 3$ and $p \leq -1$ for $T = 5$.

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for TGV

First we present the results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha^0)$ as $\alpha \to 0$ for the Taylor Green Vortex. The results show that both trends are heading to zero. This also demonstrates why we need the analysis of the slope. It does not clearly indicate from this graph alone whether or not the trend remains positive.

Results of $\boldsymbol{\alpha}$ slope for TGV

Results of p slope as $\alpha \to 0$ for the Taylor Green Vortex.

Comparison to Larios et al. 2018

Compare our results to Larios et al. 2018. Our results conflict with those of Larios et al. 2018. Their analysis found that blow-up does occur for the Taylor Green Vortex with a $p \leq -1$. Additionally the minimum slope occurs at $T^* \approx 4$ which coincides with those of Bustamante and Brachet. However, we believe their analysis improperly scales the Taylor Green Vortex. RESULTS RESULTS RESULTS.

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for INIT1

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT1})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for INIT1

Results of α slope as $\alpha \to 0$

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for INIT2

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT2})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for INIT2

Results of α slope as $\alpha \to 0$

4.0.2 Optimization Results

We present the Optimization Results for each initial condition

Results of $\boldsymbol{\alpha^2\Phi_{\alpha,T}}$ for TGV

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{TGV})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for TGV

Results of α slope as $\alpha \to 0$

Results of $\boldsymbol{\alpha^2 \Phi_{\alpha,T}}$ for INIT1

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT1})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for INIT1

Results of α slope as $\alpha \to 0$

Results of $\boldsymbol{\alpha^2 \Phi_{\alpha,T}}$ for INIT2

Results of $\alpha^2\Phi_{\alpha,T}(\boldsymbol{\eta}_{INIT2})$ as $\alpha \to 0$

Results of $\boldsymbol{\alpha}$ slope for INIT2

Results of α slope as $\alpha \to 0$

5 Conclusion

Every thesis also needs a concluding chapter

A Derivation of the Gateaux Directional Differential

The equation for the Gateaux differential (2.34) is

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') := \lim_{\epsilon\to 0}\frac{1}{\epsilon}[\Phi_{\alpha,T}(\boldsymbol{\eta}+\epsilon\boldsymbol{\eta}') - \Phi_{\alpha,T}(\boldsymbol{\eta})].$$

The objective functional (2.30) is defined as,

$$\Phi_{\alpha,T}(\boldsymbol{\eta}) := \|\nabla \boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{L^2} = \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{\dot{H}^1}.$$

Substituting the second definition of (2.30) into (2.34),

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[\|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}+\epsilon\boldsymbol{\eta}')\|^2_{\dot{H}^1} - \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta})\|^2_{\dot{H}^1}].$$

The equation for the $\dot{H}^1$ seminorm is

$$\|u\|^2_{\dot{H}^1} = (2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2.$$

Substituting in the equation for the $\dot{H}^1$ seminorm

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_{\boldsymbol{k}}+\epsilon\hat{\boldsymbol{u}}'_{\boldsymbol{k}}|^2-(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_{\boldsymbol{k}}|^2].$$

This simplifies to

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \lim_{\epsilon\to 0}\frac{1}{\epsilon}[(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2 2\epsilon|\hat{\boldsymbol{u}}_{\boldsymbol{k}}\hat{\boldsymbol{u}}'_{\boldsymbol{k}}|+(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2\epsilon^2|\hat{\boldsymbol{u}}'_{\boldsymbol{k}}|^2].$$

Taking the limit $\epsilon\to 0$

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2(2\pi)^3\sum_{k\in\mathbb{Z}^3}|k|^2|\hat{\boldsymbol{u}}_{\boldsymbol{k}}\hat{\boldsymbol{u}}'_{\boldsymbol{k}}|.$$

This can be expressed in the integral form

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\int_{\mathbb{T}^3}\sum_{k\in\mathbb{Z}^3}|k|^2\hat{\boldsymbol{u}}_{\boldsymbol{k}}\hat{\boldsymbol{u}}'_{\boldsymbol{k}}e^{2ikx}dx.$$

Which is equivalent to the inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle\sum_{k\in\mathbb{Z}^3}|k|\hat{\boldsymbol{u}}_{\boldsymbol{k}}e^{ikx},\sum_{k\in\mathbb{Z}^3}|k|\hat{\boldsymbol{u}}'_{\boldsymbol{k}}e^{ikx}\right\rangle_{L^2}.$$

This gives us the Gateaux differential in the form of the $\dot{H}^1$ inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle\boldsymbol{u}_\alpha(\cdot,T),\boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1}.$$

The integral form of the $\dot{H}^1$ inner product is

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle_{\dot{H}^1} = \int_{\mathbb{T}^3} |D|\boldsymbol{u} \cdot |D|\boldsymbol{v} \, dx,$$

where $|D|$ is the operator defined in (1.21). Therefore, the Gateaux differential can be expressed as

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\int_{\mathbb{T}^3} |D|\boldsymbol{u}_\alpha(x,T) \cdot |D|\boldsymbol{u}'_\alpha(x,T) \, dx,$$

which can be rearranged to the form

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \int_{\mathbb{T}^3} 2|D|^2\boldsymbol{u}_\alpha(x,T) \cdot \boldsymbol{u}'_\alpha(x,T) \, dx.$$

We now make use of the adjoint system terminal condition derived in Appendix (C)

$$\boldsymbol{u}^*_\alpha(x,T) = 2|D|^2\boldsymbol{u}_\alpha(x,T).$$

Therefore

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \int_{\mathbb{T}^3} \boldsymbol{u}^*_\alpha(x,T) \cdot \boldsymbol{u}'_\alpha(x,T) \, dx.$$

Now using the duality-pairing defined in Appendix (C)

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \int_{\mathbb{T}^3} \boldsymbol{\eta}'(x) \cdot \boldsymbol{u}^*_\alpha(x,0) \, dx.$$

This is the inner product

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = \langle \boldsymbol{u}^*_\alpha(x,0), \boldsymbol{\eta}'(x) \rangle_{L^2}.$$

Therefore,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\cdot,T), \boldsymbol{u}'_\alpha(\cdot,T)\right\rangle_{\dot{H}^1} = \left\langle \boldsymbol{u}^*_\alpha(x,0), \boldsymbol{\eta}'(x)\right\rangle_{L^2}.$$

B Linearization of the Euler-Voigt System

Euler-Voigt Equations,

$$\begin{cases} -\alpha^2 \partial_t \Delta \boldsymbol{u}_\alpha + \partial_t \boldsymbol{u}_\alpha + (\boldsymbol{u}_\alpha \cdot \boldsymbol{\nabla}) \boldsymbol{u}_\alpha + \boldsymbol{\nabla} p = 0 \\ \boldsymbol{\nabla} \cdot \boldsymbol{u}_\alpha = 0 \\ \boldsymbol{u}_\alpha|_{t=0} = \boldsymbol{\eta}. \end{cases}$$

Substituting in $\boldsymbol{u}_\alpha = \boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha$, $\boldsymbol{\eta} = \boldsymbol{\eta} + \boldsymbol{\eta}'$, and $p = p + p'$,

$$\begin{cases} -\alpha^2 \partial_t \Delta (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) + \partial_t (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) + ((\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) \cdot \boldsymbol{\nabla})(\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) + \boldsymbol{\nabla}(p + p') = 0 \\ \boldsymbol{\nabla} \cdot (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) = 0 \\ (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha)|_{t=0} = \boldsymbol{\eta} + \boldsymbol{\eta}'. \end{cases}$$

Expanding

$$\begin{cases} -\alpha^2\partial_t\Delta\boldsymbol{u}_\alpha - \alpha^2\partial_t\Delta\boldsymbol{u}'_\alpha + \partial_t\boldsymbol{u}_\alpha + \partial_t\boldsymbol{u}'_\alpha + \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \\ \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{\nabla}p + \boldsymbol{\nabla}p' = 0 \\ \boldsymbol{\nabla}\cdot(\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha) = 0 \\ (\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha)|_{t=0} = \boldsymbol{\eta} + \boldsymbol{\eta}'. \end{cases}$$

Substituting in the Euler-Voigt equations we simplify to

$$\begin{cases} -\alpha^2\partial_t\Delta\boldsymbol{u}'_\alpha + \partial_t\boldsymbol{u}'_\alpha + \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{\nabla}p' = 0 \\ \boldsymbol{\nabla}\cdot\boldsymbol{u}'_\alpha = 0 \\ \boldsymbol{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'. \end{cases}$$

Linearizing the system by removing non-linear terms, we derive the Linearized Euler-Voigt system

$$\begin{cases} -\alpha^2\partial_t\Delta\boldsymbol{u}'_\alpha + \partial_t\boldsymbol{u}'_\alpha + \boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p' = 0 \\ \boldsymbol{\nabla}\cdot\boldsymbol{u}'_\alpha = 0 \\ \boldsymbol{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'. \end{cases}$$

This can be rearranged to the form

$$\begin{cases} \partial_t\boldsymbol{u}'_\alpha + (I - \alpha^2\Delta)^{-1}(\boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha\cdot\boldsymbol{\nabla}\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p') = 0 \\ \boldsymbol{\nabla}\cdot\boldsymbol{u}'_\alpha = 0 \\ \boldsymbol{u}'_\alpha|_{t=0} = \boldsymbol{\eta}'. \end{cases} \tag{B.55}$$

C Derivation of the Adjoint System

For this derivation we will exclude the α subscript on the u_α, u'_α, and u^*_α terms for the sake of simplicity. Additionally, we substitute the integration with the following notation

$$\int_0^T \int_{\mathbb{T}^3} u\, dt = \int_D u\, dt$$

We derive the adjoint system using the following duality pairing relation,

$$(L[u'], u^*) := \int_D L[u']u^* dxdt = (u', L[u^*]) + \int_a^b u'u^*|_{t=T} - \int_a^b u'u^*|_{t=0} dx = 0.$$

Substituting in the equation from the Euler-Voigt perturbation system,

$$(L[u'], u^*) := \int_D (\partial_t u' + (I - \alpha^2 \Delta)^{-1}(u.\nabla u' + u' \cdot \nabla u_+ \nabla p'))u^* + (\nabla \cdot u')p^* dx\, dt.$$

This integral can be split into five parts. Solving the first part with integration by parts, with $u = u^*$ and $v' = \partial_t u'$

$$\int_D u^* \partial_t u' dx\, dt = \int_{\mathbb{T}^3} u^* u'|_{t=0}^{t=T} dx - \int_D u' \partial_t u^* dx\, dt.$$

Starting with the second part and expressing it in Einstein Summation notation,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = \left[(I-\alpha^2\Delta)^{-1}u_j\frac{\partial u_i'}{\partial x_j}\right]u_i^* dx\,dt.$$

Using the derivative product rule and the fact that $\nabla\cdot u = 0$,

$$\frac{\partial}{\partial x_j}(u_j u_i') = \frac{\partial u_j}{\partial x_j}u_i' + u_j\frac{\partial u_i'}{\partial x_j} = u_j\frac{\partial u_i'}{\partial x_j}.$$

Substituting this into the integral,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = \left[(I-\alpha^2\Delta)^{-1}\frac{\partial}{\partial x_j}(u_j u_i')\right]u_i^* dx\,dt.$$

Using the periodic boundaries of the system, we can now apply integration by parts with $u = [(I-\alpha^2\Delta)^{-1}u_i^*]$ and $v' = \frac{\partial}{\partial x_j}(u_j u_i')$,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = -u_j u_i'\frac{\partial}{\partial x_j}\left[(I-\alpha^2\Delta)^{-1}u_i^*\right]dx\,dt.$$

Converting back to vector notation,

$$\left[(I-\alpha^2\Delta)^{-1}u\cdot\nabla u'\right]\cdot u^* dx\,dt = -u'\cdot\left[u\cdot\nabla\left[(I-\alpha^2\Delta)^{-1}u^*\right]\right]dx\,dt.$$

Starting with the third part and expressing it in Einstein Summation notation,

$$\left[(I-\alpha^2\Delta)^{-1}u'\cdot\nabla u\right]\cdot u^* dx\,dt = \left[(I-\alpha^2\Delta)^{-1}u_j'\frac{\partial u_i}{\partial x_j}\right]u_i^* dx\,dt.$$

Applying integration by parts with $u = [(I - \alpha^2\Delta)^{-1}u_i^*]$ and $v' = \frac{\partial}{\partial x_j}(u_j' u_i)$

$$\left[(I - \alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -u_j' u_i \frac{\partial}{\partial x_j}\left[(I - \alpha^2\Delta)^{-1}u_i^*\right] dx\, dt.$$

Because the goal is to extract u' and $v \cdot w = v^T \cdot w^T$, we can apply a transpose to the above equation and so swap the indices to,

$$\left[(I - \alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -u_i' u_j \frac{\partial}{\partial x_i}\left[(I - \alpha^2\Delta)^{-1}u_j^*\right] dx\, dt.$$

Converting back to vector notation,

$$\left[(I - \alpha^2\Delta)^{-1}u' \cdot \nabla u\right] \cdot u^* dx\, dt = -u' \cdot \left[u \cdot \left[\nabla (I - \alpha^2\Delta)^{-1}u^*\right]^T\right] dx\, dt.$$

Starting with the fourth part of the integral and expressing it in Einstein Summation notation,

$$\nabla p' \cdot u^* dx\, dt = \frac{\partial p'}{\partial x_i} u_i^* dx\, dt.$$

Applying integration by parts with $u = u_i^*$ and $v' = \partial x_i p'$,

$$\nabla p' \cdot u^* dx\, dt = -p' \frac{\partial u_i^*}{\partial x_i} dx\, dt.$$

Converting back to vector notation,

$$\nabla p' \cdot u^* dx\, dt = -p'(\nabla \cdot u^*) dx\, dt.$$

Starting with the final part of the integral and expressing it in Einstein Summation notation,

$$(\nabla \cdot u')p^* dx\, dt = \frac{\partial u_i'}{\partial x_i} p^* dx\, dt.$$

Applying integration by parts with $u = p^*$ and $v' = \partial x_i u_i'$,

$$(\nabla \cdot u')p^* dx\, dt = -\frac{\partial p^*}{\partial x_i} u_i' dx\, dt.$$

Converting back to vector notation,

$$(\nabla \cdot u')p^* dx\, dt = -u' \cdot \nabla p^* dx\, dt.$$

Combining the integrals,

$$\int_{\mathbb{T}^3} u^* u'|_{t=0}^{t=T} dx - u' \left[\partial_t u^* - \left[u \cdot \nabla \left[(I - \alpha^2 \Delta)^{-1} u^*\right]\right] - \left[u \cdot \left[\nabla (I - \alpha^2 \Delta)^{-1} u^*\right]^T\right] - \nabla p^*\right] - p'(\nabla \cdot u^*)$$

This gives us our adjoint system,

$$\mathbb{L}^* \begin{bmatrix} u_\alpha^* \\ \\ p_\alpha^* \end{bmatrix} := \begin{bmatrix} -\partial_t u^* - (\nabla (I - \alpha^2)^{-1} u_\alpha^* + (\nabla (I - \alpha^2)^{-1} u_\alpha^*)^T) u_\alpha - \nabla p_\alpha^* \\ \\ -\nabla \cdot u_\alpha^* \end{bmatrix} = \begin{bmatrix} 0 \\ \\ 0 \end{bmatrix}$$

To derive the terminal condition of the adjoint system we compare the second term of the duality pairing to the objective functional.

$$\int_{\mathbb{T}^3} u'(x,T) \boldsymbol{u}^*(x,T) dx = 2 \left\langle u(x,T), u'(x,T) \right\rangle_{\dot{H}^1}$$

$$= 2\int_{\mathbb{T}^3} |D|u(x,T)\boldsymbol{u}'(x,T)dx$$

$$= \int_{\mathbb{T}^3} 2|D|^2u(x,T)\boldsymbol{u}'(x,T)dx$$

Therefore,

$$\boldsymbol{u}^*(x,T) = 2|D|^2u(x,T)$$

D Proof of ~~Gevrey Class size~~

Proof. The definition of the Gevrey class (1.19) is

$$G^\sigma := \left\{ \boldsymbol{v} \in H^s(\mathbb{T}^3) : \|\boldsymbol{v}\|_{G^\sigma} = \langle \boldsymbol{v}, \boldsymbol{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0.$$

The definition of the Gevrey inner product (1.20) is

$$\forall \boldsymbol{v}, \boldsymbol{u} \in G^\sigma, \quad \langle \boldsymbol{v}, \boldsymbol{u} \rangle_{G^\sigma} := \int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma |D|} \boldsymbol{v} \cdot \boldsymbol{u} \, d\boldsymbol{x}.$$

Let σ_1, σ_2 be defined such that

$$0 < \sigma_1 < \sigma_2.$$

Therefore,

$$\begin{aligned}
\|\boldsymbol{v}\|_{G^{\sigma_2}} &= \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma_2 |D|} \boldsymbol{v} \cdot \boldsymbol{v} \, d\boldsymbol{x} \right]^{1/2} \\
&> \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s e^{2\sigma_1 |D|} \boldsymbol{v} \cdot \boldsymbol{v} \, d\boldsymbol{x} \right]^{1/2} = \|v\|_{G^{\sigma_1}} \\
&> \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s e^{0} \boldsymbol{v} \cdot \boldsymbol{v} \, d\boldsymbol{x} \right]^{1/2} = \|\boldsymbol{v}\|_{G^0} \\
&= \left[\int_{\mathbb{T}^3} (1 + |D|^2)^s \boldsymbol{v} \cdot \boldsymbol{v} \, d\boldsymbol{x} \right]^{1/2} = \|v\|_{H^s} \qquad \text{(D.56)}
\end{aligned}$$

Thus,

$$\|\boldsymbol{v}\|_{G^{\sigma_2}} > \|\boldsymbol{v}\|_{G^{\sigma_1}} > \|\boldsymbol{v}\|_{G^0}.$$

It follows from this that if $\|\boldsymbol{v}\|_{G^{\sigma_2}} < \infty$ then $\|\boldsymbol{v}\|_{G^{\sigma_1}} < \infty$ and so if $v \in G^{\sigma_2}$ then $v \in G^{\sigma_1}$. ~~However it does not follow from this that if~~ $\cancel{\|\boldsymbol{v}\|_{G^{\sigma_1}} < \infty}$ ~~then~~ $\cancel{\|\boldsymbol{v}\|_{G^{\sigma_2}} < \infty}$~~, and so it is not required that if~~ $\cancel{v \in G^{\sigma_1}}$ ~~then~~ $\cancel{v \in G^{\sigma_2}}$. The same argument shows that if $v \in G^{\sigma_1}$ then $v \in G^0$ which is equivalent to $v \in H^s$. Therefore,

$$G^{\sigma_2} \subset G^{\sigma_1} \subset G^0 = H^s$$

□

E $\|\boldsymbol{\eta}\|_{\dot{H}^1}$ for the General Taylor Green Vortex

The general form of the 3D Taylor Green Vortex on the 3D torus Ω with characteristic length L is

$$\cancel{\boldsymbol{\eta}_{TGV}} = \begin{bmatrix} \cancel{V_0 \sin(2\pi x/L) \cos(2\pi y/L) \cos(2\pi z/L)} \\ \cancel{-V_0 \cos(2\pi x/L) \sin(2\pi y/L) \cos(2\pi z/L)} \\ \cancel{0.} \end{bmatrix}$$

To simplify the notation we drop the subscript TGV and substitute in

$$x' = 2\pi x/L, \quad y' = 2\pi y/L, \quad z' = 2\pi z/L.$$

The formula for $\|\boldsymbol{\eta}\|_{\dot{H}^1}$ is

$$\|\boldsymbol{\eta}\|_{\dot{H}^1} = \|\nabla\boldsymbol{\eta}\|_{L^2} = \left[\sum_{i,j=1}^{3} \int_{\Omega} (\partial_i \boldsymbol{\eta}_j)^2 \, d\boldsymbol{x}\right]^{1/2},$$

where ∂_i is the partial derivative with respect to the i-th component and $\boldsymbol{\eta}_j$ is the j-th component of $\boldsymbol{\eta}$. Because the 3rd component of $\boldsymbol{\eta}$ is zero, the partial derivative is also zero. This leaves us with six integrals to evaluate.

$$\begin{aligned}\sum_{i,j=1}^{3}\int_\Omega (\partial_i\boldsymbol{\eta}_j)^2\,d\boldsymbol{x} &= \int_\Omega (\partial_1\boldsymbol{\eta}_1)^2\,d\boldsymbol{x} + \int_\Omega (\partial_2\boldsymbol{\eta}_1)^2\,d\boldsymbol{x} + \int_\Omega (\partial_3\boldsymbol{\eta}_1)^2\,d\boldsymbol{x}\\ &\quad + \int_\Omega (\partial_1\boldsymbol{\eta}_2)^2\,d\boldsymbol{x} + \int_\Omega (\partial_2\boldsymbol{\eta}_2)^2\,d\boldsymbol{x} + \int_\Omega (\partial_3\boldsymbol{\eta}_2)^2\,d\boldsymbol{x}.\end{aligned}$$

Evaluating each partial derivative

$$\begin{aligned}\sum_{i,j=1}^{3}\int_\Omega (\partial_i\boldsymbol{\eta}_j)^2\,d\boldsymbol{x} &= \int_\Omega (2\pi V_0/L\cos(x')\cos(y')\cos(z'))^2\,d\boldsymbol{x}\\ &\quad + \int_\Omega (-2\pi V_0/L\sin(x')\sin(y')\cos(z'))^2\,d\boldsymbol{x}\\ &\quad + \int_\Omega (-2\pi V_0/L\sin(x')\cos(y')\sin(z'))^2\,d\boldsymbol{x}\\ &\quad + \int_\Omega (2\pi V_0/L\sin(x')\sin(y')\cos(z'))^2\,d\boldsymbol{x}\\ &\quad + \int_\Omega (-2\pi V_0/L\cos(x')\cos(y')\cos(z'))^2\,d\boldsymbol{x}\\ &\quad + \int_\Omega (2\pi V_0/L\cos(x')\sin(y')\sin(z'))^2\,d\boldsymbol{x}\cancel{.}\end{aligned}$$

~~Which~~ simplifies to

$$\begin{aligned}\sum_{i,j=1}^{3}\int_\Omega (\partial_i\boldsymbol{\eta}_j)^2\,d\boldsymbol{x} &= 4\pi^2V_0^2/L^2\int_\Omega \cos^2(x')\cos^2(y')\cos^2(z')\,d\boldsymbol{x}\\ &\quad + 4\pi^2V_0^2/L^2\int_\Omega \sin^2(x')\sin^2(y')\cos^2(z')\,d\boldsymbol{x}\\ &\quad + 4\pi^2V_0^2/L^2\int_\Omega \sin^2(x')\cos^2(y')\sin^2(z')\,d\boldsymbol{x}\\ &\quad + 4\pi^2V_0^2/L^2\int_\Omega \sin^2(x')\sin^2(y')\cos^2(z')\,d\boldsymbol{x}\\ &\quad + 4\pi^2V_0^2/L^2\int_\Omega \cos^2(x')\cos^2(y')\cos^2(z')\,d\boldsymbol{x}\\ &\quad + 4\pi^2V_0^2/L^2\int_\Omega \cos^2(x')\sin^2(y')\sin^2(z')\,d\boldsymbol{x}.\end{aligned}$$

Because the integrand is separable, the integrals can be reduced to products of,

$$\int_0^L \cos^2(x')\,dx = \frac{L}{2}, \qquad \text{and} \qquad \int_0^L \sin^2(x')\,dx = \frac{L}{2}.$$

Therefore, each of the six integrals is equal to

$$\frac{4\pi^2 V_0^2}{L^2}\frac{L^3}{8}.$$

The sum of the six integrals is

$$\sum_{i,j=1}^{3} \int_\Omega (\partial_i \boldsymbol{\eta}_j)^2\,d\boldsymbol{x} = 6\frac{4\pi^2 V_0^2}{L^2}\frac{L^3}{8} = 3\pi^2 V_0^2 L.$$

Substituting into the formula for $\|\cdot\|_{\dot{H}^1}$

$$\|\boldsymbol{\eta}_{TGV}\|_{\dot{H}^1} = \left[3\pi^2 V_0^2 L\right]^{1/2} = \sqrt{3L}\pi V_0.$$

F $\|\nabla \times \boldsymbol{\eta}\|_{L^\infty}$ for the General Taylor Green Vortex

The general form of the 3D Taylor Green Vortex on the 3D torus Ω with characteristic length L is

$$\boldsymbol{\eta}_{TGV} = \begin{bmatrix} V_0 \sin(2\pi x/L)\cos(2\pi y/L)\cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L)\sin(2\pi y/L)\cos(2\pi z/L) \\ 0. \end{bmatrix}$$

To simplify the notation we drop the subscript TGV and substitute in

$$x' = 2\pi x/L, \quad y' = 2\pi y/L, \quad z' = 2\pi z/L.$$

The vorticity of $\boldsymbol{\eta}$ is

$$\nabla \times \boldsymbol{\eta} = \begin{bmatrix} \partial_2 \boldsymbol{\eta}_3 - \partial_3 \boldsymbol{\eta}_2 \\ \partial_3 \boldsymbol{\eta}_1 - \partial_1 \boldsymbol{\eta}_3 \\ \partial_1 \boldsymbol{\eta}_2 - \partial_2 \boldsymbol{\eta}_1. \end{bmatrix}$$

where ∂_i is the partial derivative with respect to the i-th component and $\boldsymbol{\eta}_j$ is the j-th component of $\boldsymbol{\eta}$. Substituting in the partial derivatives

$$\nabla \times \boldsymbol{\eta} = \begin{bmatrix} 0 - (-(2\pi V_0/L)\cos(x')\sin(y')\sin(z') \\ -(2\pi V_0/L)\sin(x')\cos(y')\sin(z') - 0 \\ (2\pi V_0/L)\sin(x')\sin(y')\cos(z') + (2\pi V_0)\sin(x')\sin(y')\cos(z'). \end{bmatrix}$$

which simplifies to

$$\nabla \times \eta_{TG} = \begin{bmatrix} (2\pi V_0/L)\cos(x')\sin(y')\sin(z') \\ -(2\pi V_0/L)\sin(x')\cos(y')\sin(z') \\ (4\pi V_0/L)\sin(x')\sin(y')\cos(z') \end{bmatrix}$$

The $\|\nabla \times \boldsymbol{\eta}\|_{L^\infty}$ norm is the maximum absolute value of the components of $\nabla \times \boldsymbol{\eta}$.

$$\begin{aligned} \|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = \max(&|(2\pi V_0/L)\cos(x')\sin(y')\sin(z')| \\ &, |-(2\pi V_0/L)\sin(x')\cos(y')\sin(z')| \\ &, |(4\pi V_0/L)\sin(x')\sin(y')\cos(z')|) \end{aligned}$$

The maximums are

$$\|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = \max((2\pi V_0/L), (2\pi V_0/L), (4\pi V_0/L))$$

Therefore,

$$\|\nabla \times \boldsymbol{\eta}_{TGV}\|_{L^\infty} = 4\pi V_0/L.$$

G Rescaling the Euler equations

For Euler fluids we have the equation

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \boldsymbol{\nabla})\boldsymbol{u} + \boldsymbol{\nabla} p = 0,$$

where the solution is $\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x}, t)$. We construct a new scaled solution with

$$\tau = \lambda t, \qquad \text{and} \qquad \boldsymbol{y} = \beta \boldsymbol{x}.$$

where $\lambda, \beta > 0$ are constants. Our new solution $\boldsymbol{v}$ is defined as

$$\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{y}, \tau) = \gamma \boldsymbol{u}(\boldsymbol{y}/\beta, \tau/\lambda).$$

Therefore

$$\partial_t \boldsymbol{u} = \frac{1}{\gamma} \frac{\partial \boldsymbol{v}}{d\tau} \frac{d\tau}{dt} = \frac{\lambda}{\gamma} \partial_\tau \boldsymbol{v}.$$

and

$$\boldsymbol{\nabla} \boldsymbol{u} = \frac{1}{\gamma} \boldsymbol{\nabla} v \frac{d\boldsymbol{y}}{d\boldsymbol{x}} = \frac{\beta}{\gamma} \boldsymbol{\nabla} v.$$

Substituting into the original equation

$$\frac{\lambda}{\gamma}\partial_\tau \boldsymbol{v} + \frac{\beta}{\gamma^2}(\boldsymbol{v}\cdot\boldsymbol{\nabla})\boldsymbol{v} + \boldsymbol{\nabla} q = 0,$$

where q is defined as whatever pressure is required to satisfy the equality. This can be rearranged to the form,

$$\frac{\lambda\gamma}{\beta}\partial_\tau \boldsymbol{v} + (\boldsymbol{v}\cdot\boldsymbol{\nabla})\boldsymbol{v} + \boldsymbol{\nabla} q' = 0.$$

Therefore, our function $\boldsymbol{v}$ can only be a solution to the Euler equations if

$$\frac{\lambda\gamma}{\beta} = 1.$$

For the Taylor Green Vortex with $L = 2\pi$ and $V_0 = 1$, let

$$\gamma = \lambda = \beta = 1.$$

In order to rescale this to the domain with $L = 1$ we must set $\beta = 1/2\pi$. Additionally we want our rescaled solution to have the same time scale as the original solution and so we must set $\lambda = 1$. Therefore, for our scaled solution $\boldsymbol{v}$ to be a solution of Euler we must set $\gamma = V_0 = 1/2\pi$.

H Rescaling the Euler-Voigt equations

For Euler-Voigt fluids we have the equation

$$-\alpha^2\partial_t\boldsymbol{\Delta}\boldsymbol{u}_\alpha + \partial_t\boldsymbol{u}_\alpha + (\boldsymbol{u}_\alpha\cdot\boldsymbol{\nabla})\boldsymbol{u}_\alpha + \boldsymbol{\nabla}p = 0,$$

where the solution is $\boldsymbol{u}_\alpha = \boldsymbol{u}_\alpha(\boldsymbol{x}, t)$. We construct a new scaled solution with

$$\tau = \lambda t, \qquad \kappa = \beta\alpha, \qquad \boldsymbol{y} = \beta\boldsymbol{x}.$$

where $\lambda, \beta > 0$ are constants. The variables $\boldsymbol{x}$ and α are both scaled by β because they are both spatial and we want to keep the relative size of the filter to the domain size fixed. Our new solution $\boldsymbol{v}$ is defined as

$$\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{y}, \tau) = \gamma\boldsymbol{u}_\alpha(\boldsymbol{y}/\beta, \tau/\lambda).$$

Therefore

$$\partial_t\boldsymbol{u}_\alpha = \frac{1}{\gamma}\frac{\partial\boldsymbol{v}}{d\tau}\frac{d\tau}{dt} = \frac{\lambda}{\gamma}\partial_\tau\boldsymbol{v}.$$

and

$$\boldsymbol{\nabla}\boldsymbol{u}_\alpha = \frac{1}{\gamma}\boldsymbol{\nabla}v\frac{d\boldsymbol{y}}{d\boldsymbol{x}} = \frac{\beta}{\gamma}\boldsymbol{\nabla}\boldsymbol{v}.$$

and

$$\partial_t \boldsymbol{\Delta u}_\alpha = \partial_t \frac{\beta^2}{\gamma} \boldsymbol{\Delta v} = \frac{\lambda\beta^2}{\gamma} \partial_\tau \boldsymbol{\Delta v}.$$

Substituting into the original equation

$$-\alpha^2 \frac{\lambda\beta^2}{\gamma} \partial_\tau \boldsymbol{\Delta v} + \frac{\lambda}{\gamma} \partial_\tau \boldsymbol{v} + \frac{\beta}{\gamma^2} (\boldsymbol{v} \cdot \boldsymbol{\nabla}) \boldsymbol{v} + \boldsymbol{\nabla} q = 0,$$

where q is defined as whatever pressure is required to satisfy the equality. This can be rearranged to the form,

$$-\kappa^2 \partial_t \boldsymbol{\Delta v} + \partial_\tau \boldsymbol{v} + \frac{\beta}{\lambda\gamma} (\boldsymbol{v} \cdot \boldsymbol{\nabla}) \boldsymbol{v} + \boldsymbol{\nabla} q' = 0.$$

Therefore, our function $\boldsymbol{v}$ can only be a solution to the Euler-Voigt equations if

$$\frac{\beta}{\lambda\gamma} = 1.$$

For the Taylor Green Vortex with $L = 2\pi$ and $V_0 = 1$, let $\gamma = \lambda = \beta = 1$. In order to rescale this to the domain with $L = 1$ we must set $\beta = 1/2\pi$. Additionally we want our rescaled solution to have the same time scale as the original solution and so we must set $\lambda = 1$. Therefore, for our scaled solution $\boldsymbol{v}$ to be a solution of Euler-Voigt we must set $\gamma = V_0 = 1/2\pi$.

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